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Climate change estimation and prediction over eastern Afromontane biodiversity hotspot areas

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ABSTRACT

The Eastern Afromontane Biodiversity Hotspot Area (EABHA) is among the world's most biologically rich regions. Yet these highland systems are increasingly threatened by climate change. In this study, we assess long-term climate variability and project future trends to 2050, aligning our analysis with the United Nations Sustainable Development Goals (SDGs). Using Principal Component Analysis (PCA), Empirical Orthogonal Function (EOF), and an artificial neural network model (ANNM), we examined precipitation, temperature, rainy-day frequency, and wind dynamics from satellite observations, in-situ measurements, and reanalysis climate models. The ANNM was trained on detrended EOF components and station data (1981–2022) and validated using a data split into training (80%), validation (10%), and test (10%) sets, ensuring representation across regions and time periods. Results reveal a marked decline in the number of rainy days over the past four decades, a trend expected to persist through 2050. The annual mean temperature has increased and is projected to rise by ~0.040 K yr^{-1} for 2023–2050 relative to the 1981–2022 baseline (0.030 K yr^{-}). Reductions in the number of rainy days are the primary drivers of drought events in ecosystems located above 2800 m a.s.l. in the study area. These findings advance SDG 13 (Climate Action) by quantifying climate risks, SDG 15 (Life on Land) by identifying threats to biodiversity, and SDG 6 (Clean Water and Sanitation) by highlighting freshwater security challenges. By demonstrating that drought risk is more closely linked to declining rain-day frequency than annual rainfall totals, this research provides actionable insights for adaptation, conservation, and resilience planning across the EABHA.

1. Introduction

The highland ecosystems of the Afromontane Floristic Region are ecologically distinctive, characterized by an "archipelago-like" pattern of biodiversity across Africa's mountain ranges. The eastern portion of this region, designated as the Eastern Afromontane Biodiversity Hotspot Area (EABHA), spans parts of the Arabian Peninsula, Ethiopia, Eritrea, Sudan, Djibouti, Kenya, and Somalia (Razgour et al., 2020). Celebrated for its exceptional species richness and high levels of endemism, the EABHA provides critical ecosystem services, including climate regulation and freshwater provisioning for millions of people. Despite its ecological significance, the region faces escalating threats from human activities, such as habitat fragmentation, land-use change, and deforestation, which contribute to climate instability and the erosion of genetic resources (Razgour et al., 2020).

Substantial forest loss and ecological degradation within the EABHA have been well documented (Dessie and Kleman, 2007; Muluneh, 2021). For example, the Ethiopian Highlands and Eastern Arc Mountains have experienced dramatic deforestation, with forest cover reduced to less than 4% of its historical extent in Ethiopia and approximately 15% in Eritrea (Dimitrov, 2010; CEPF, 2012). Similarly, wooded coffee landscapes in southwest Ethiopia lost over 67% of their forest cover between 1973 and 2005, and comparable patterns of forest decline have been observed in the Albertine Rift and the Awasa watershed (Dessie and Kleman, 2007; Muluneh, 2021). Although these studies highlight significant environmental change, prior climate research in the region, including Solomon (2007) and Muluneh (2021), often lacked a comprehensive assessment of spatial and temporal climate variability across different elevations, particularly concerning temperature fluctuations, rainy day frequency, and altitude-specific projections.

Addressing these challenges aligns closely with several United Nations Sustainable Development Goals (SDGs). This study directly contributes to SDG 13 (Climate Action) by examining climate variability and change, SDG 15 (Life on Land) by highlighting forest and

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biodiversity loss, and SDG 6 (Clean Water and Sanitation) by emphasizing ecosystem services that support water security. Given that the alpine ecosystems of the EABHA underpin the livelihoods of millions, SDG 1 (No Poverty) and SDG 2 (Zero Hunger) are also highly relevant. To bridge existing research gaps and support sustainable management, this study conducts a comprehensive spatiotemporal analysis of climate change and variability across the EABHA.

The comparison of climate trends above and below 2800 m a.s.l. is grounded in the well-established concept of elevation-dependent climate change (EDCC), whereby high-altitude regions often experience amplified warming and distinct hydroclimatic responses relative to lowlands (Rangwala and Miller, 2012; Pepin et al., 2015). In tropical and subtropical mountain systems, including the EABHA, elevations around ~2500–3000 m frequently represent ecological and climatic transition zones, such as the upper montane forest–afroalpine boundary, where temperature sensitivity, cloud formation, and moisture dynamics change markedly (Palazzi et al., 2019; You et al., 2020). Recent assessments, including the IPCC AR6, indicate that high-elevation environments are particularly sensitive to reductions in cloud cover, changes in atmospheric lapse rates, and shifts in precipitation characteristics, especially a decline in rainfall frequency rather than total accumulation (Pepin et al., 2015; IPCC, 2021). Furthermore, mountain regions exhibit strong coupling between rising temperature and decreasing rainy day frequency, which can intensify drought conditions and ecosystem stress despite relatively stable annual precipitation totals (Rangwala and Miller, 2012; Palazzi et al., 2019). Therefore, stratifying the analysis at 2800 m a.s.l. provides a physically and ecologically meaningful threshold to capture nonlinear altitude-dependent climate responses. In this regard, this paper is aimed at climate change estimation and prediction over eastern Afromontane biodiversity hotspot areas to inform science-based policies and guide conservation, adaptation, and resilience-building strategies at local and regional levels, contributing to the achievement of relevant SDGs.

2. Study area, data sources, and methodology

2.1. Study area description

The EABHA is a discontinuous mountainous system extending from the Asir Mountains of southwestern Saudi Arabia and the Yemeni highlands in the north to Mozambique and Zimbabwe in the south, covering approximately 7000 km (Fig. 1). The spatial extent of the study area, including all 17 countries encompassed within the EABHA, is clearly indicated in Fig. 1 and is therefore not repeated here. Geologically, the highlands of Ethiopia and the Arabian Peninsula were formed by the uplift of ancient rocks between 55 and 75 million years ago due to mantle-driven magmatic activity, followed by extensive flood basalt coverage around 30 million years ago. Subsequent rifting approximately 13 million years ago led to the formation of the present-day Great Rift Valley, while later volcanic activity shaped the Kenyan and Tanzanian highlands, including major peaks such as Mount Kilimanjaro, Mount Kenya, and Mount Elgon (Fjeldså and Bowie, 2021).

The region exhibits substantial altitudinal variation, ranging from lowland areas at approximately 300 m a.s.l. to alpine environments exceeding 5800 m a.s.l., contributing to its high ecological and climatic heterogeneity (Assefa, 2017). Forest ecosystems are generally distributed above 800–1000 m a.s.l., although this threshold varies regionally (CEPF, 2012). The Ethiopian Highlands alone account for a significant portion of the EABHA and are recognized for their ecological importance, including hosting the only known wild populations of Coffea arabica. Hydrologically, the region is influenced by major drainage systems such as the Congo and Nile basins, with watershed boundaries

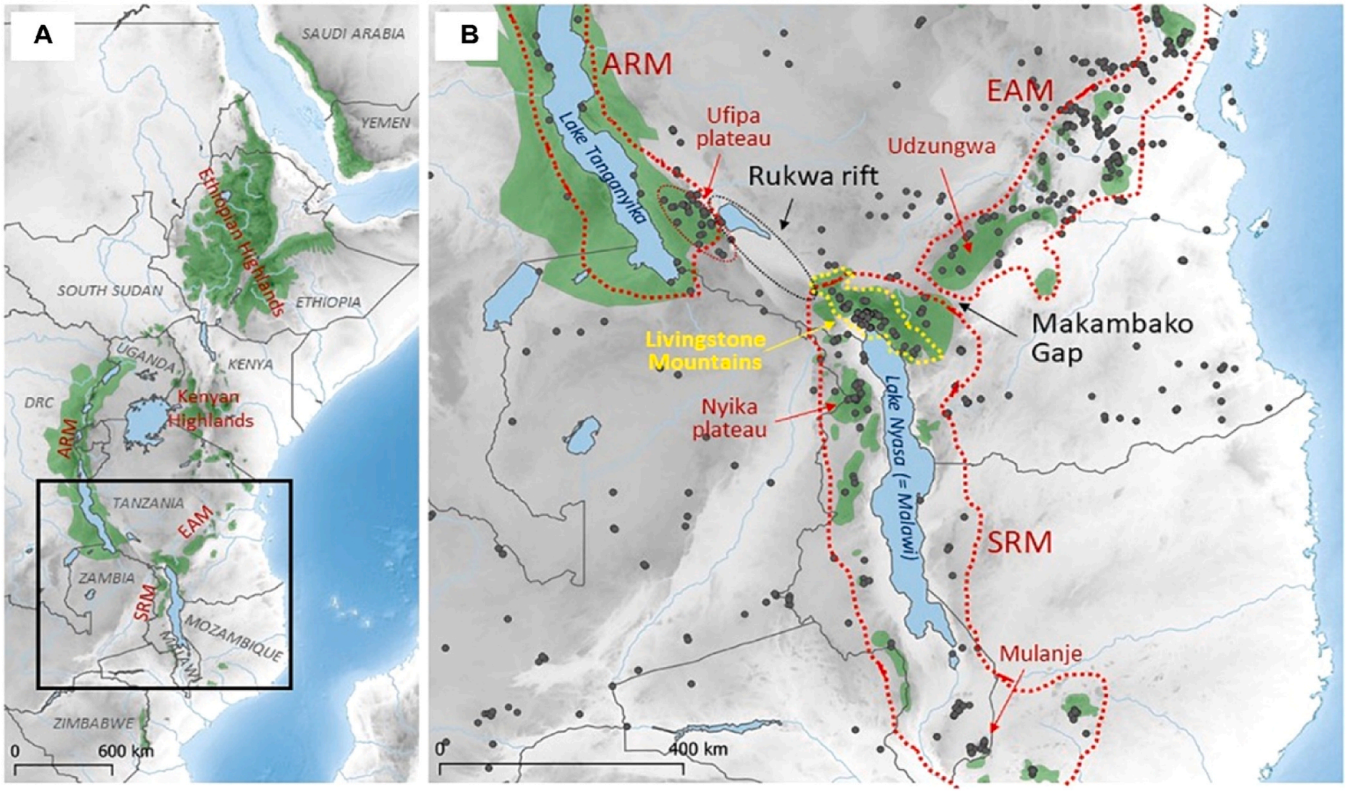


Fig. 1. Map of the EABHA (Jake et al., 2022). The grey dots indicate the locations of meteorological stations, while the box outlines the meteorological simulation domain. The red dashed line marks the boundary of the lake within the study area. The station coverage of Saudi Arabia (30), Yemen (15), Eritrea (15), Ethiopia (43), Kenya (20), Sudan (10), South Sudan (5), Somalia (10), Tanzania (30), Democratic Republic of Congo (20), Uganda (20), Burundi (5), Rwanda (15), Malawi (10), Mozambique (20), Zambia (12), and Zimbabwe (20) were used for this study.

defined by features like the Virunga Mountains (Fjeldså and Bowie, 2021). The diversity of volcanic substrates, ranging from young lava flows to highly weathered soils, further supports the region's rich biodiversity (Jake et al., 2022).

2.2. Data sources and types

Precipitation (P) and temperature (T) data were obtained from multiple meteorological stations across the EABHA through the World Meteorological Organization (WMO), covering the period 1981–2022. Due to sparse and uneven station distribution, particularly in mountainous regions, the EABHA is considered data-scarce (Fick and Hijmans, 2017). To address these limitations, additional gridded datasets from reanalysis and satellite products, including ECMWF, CRU, Afri-CORDEX RCM, and CHIRPS, were incorporated to improve spatial and temporal coverage.

- ECMWF (European Centre for Medium-Range Weather Forecasts) – daily P, T, number of rain days (NRD), zonal (U) and meridional (V) winds at 10 m, 0.125° × 0.125° spatial resolution were obtained.
- Regional Climate Models (RCM, Afri-CORDEX) – daily P and T, 0.25° × 0.25° resolution were investigated.
- Climatic Research Unit (CRU, University of East Anglia) – monthly P in 0.5° × 0.5° spatial resolution were obtained.
- CHIRPS (Climate Hazards Group InfraRed Precipitation with Station data) daily P and NRD, 0.05° × 0.05° resolution were obtained.

Detailed information regarding dataset sources, spatial and temporal resolution, and measured parameters is summarized in Table 1. These datasets were used to complement and validate in-situ observations, ensuring a robust and consistent spatiotemporal analysis of climate variables across the study area.

The different data sources website is provided in the following:

1. In-Situ: World Meteorological Organization: www.public.wmo.int/en
2. Climatic Research Unit (CRU), University of East Anglia: www.cru data.uea.ac.uk/cru/data/hrg/ (Harris et al., 2020; Morsy et al., 2022)
3. Afri-CORDEX Regional Climate Model Data (RCM): www.euro-cordex.net/ (Copernicus Climate Change Service, Climate Data Store, 2019; Yersaw and Chane, 2024)
4. Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS): www.chc.ucsb.edu/data/chirps (Ahmed et al., 2024)
5. European Center of Medium Range Weather Forecasts (ECMWF): www.cds.climate.copernicus.eu (Hersbach et al., 2020)

2.3. Methodology

The outliers in the data had been removed by applying smooth average eigenvalue statistical techniques, whereas the missing data were filled using the weighted average interpolation method. ECMWF, CRU, CHIRPS, and RCM data have different space and time resolutions that challenge comparisons. Therefore, projection and interpolation techniques were applied to convert the data to a 0.05-degree spatial resolution and daily temporal resolution using the Climate Data Operator (CDO) software in the Linux environment. After creating the different datasets to similar resolutions, validation was performed by checking performance using various error metrics, including bias ratio (BR), correlation coefficient (R), Pearson correlation (P), root mean squared error (RMSE), and mean relative error (MRE). The data was plotted and analyzed using MATLAB software. The in-situ data is used to validate the model data over EABHA. Wind data is analyzed to get the seasonal dynamics of moisture over EABHA. The Fast Fourier Transform (FFT) model was applied to remove the redundancy of the climate data for computational simplicity. The spatial variability of climate parameters was applied using the empirical orthogonal function (EOF) model, and the temporal variability of climate parameters was applied using principal component analysis (PCA).

Artificial Neural Network Model (ANNM) Climate prediction (P, T, and NRDs by altitude class based on various scenarios was analyzed up to 2050 using a feed-forward multilayer perceptron. The architecture consisted of two hidden layers (64 and 32 neurons) with ReLU activation, and an output layer predicting P, T, and NRDs. Inputs included detrended and standardized EOF principal components, altitude class (>2800 *m a.s.l.* vs <2800 *m a.s.l.*), and seasonal wind indices. Targets were observed for P, T, and NRDs from 1981 to 2022. The dataset was split 80% for training, 10% for testing, and 10% for validation. Optimization used the Adam algorithm (learning rate 0.001), mean squared error loss, and early stopping to prevent overfitting. Non-stationarity was handled by detrending and normalizing predictors before training. Projections for 2023–2050 were generated under the input value of the ANNM data P, T, NRDs, U, and V wind fields from ECMWF reanalysis sources, since, relatively, other datasets (RCM, CHIRPS, and CRU), ECMWF has the highest performance compared to ground-based stations. These variables were first statistically downscaled and temporally aligned with the historical observation period used to train the ANN. After preprocessing (standardization and normalization), the data were fed into the trained ANN as future input features. The ANN then used these predictors to generate projections of the target variable for 1981–2022 to forecast 2023–2050.

The Self-Calibrated Palmer Drought Severity Index (Sc-PDSI) was selected as the primary drought assessment metric because it integrates atmospheric and hydrological processes and automatically adjusts to local climatic conditions, improving spatial comparability in heterogeneous environments such as the EABHA. Sc-PDSI uses a water-balance model incorporating P, T, evapotranspiration (ET), recharge (R), runoff (RO), loss (L), potential evapotranspiration (PET), potential recharge (PR), potential runoff (PRO), and potential loss (PL). Palmer's duration factors (0.897 and 1/3) were applied to better represent drought persistence and recovery (Megbar and Tadesse, 2016). In addition, Sc-PDSI values were evaluated in relation to the total NRD recorded across the hotspot areas to capture how drought severity aligns with rainfall frequency patterns critical for ecosystem functioning. Alternative indices such as the Standardized Precipitation Index (SPI), the Standardized Precipitation Evapotranspiration Index (SPEI), and the original Palmer Drought Severity Index (PDSI) were less suitable for the study. The SPI relies exclusively on P and cannot account for T-driven moisture stress. The SPEI includes T effects but does not model soil-moisture or hydrological components (Megbar and Tadesse, 2016). Although the PDSI incorporates T and P, it lacks the adaptive calibration mechanism of Sc-PDSI, limiting its applicability in regions with strong climatic gradients. For these reasons, Sc-PDSI provides a more robust and ecologically relevant drought indicator for the EABHA, which analyzes the spatial orthogonality of the extreme events (drought) via EOF.

The anomaly field EOF for a special pattern and associated principal

Table 1
Details of the meteorological datasets used in this study.

Data Type	Data Used	Temporal Resolution	Spatial Resolution	Parameters
in-situ	1981–2022	Daily	300 stations across EABHA	Daily precipitation (P) and temperature (T)
CRU	1981–2022	Monthly	0.5° × 0.5°	daily P
RCM	1981–2022	Daily	0.25° × 0.25°	daily P
CHIRPS	1981–2022	Daily	0.05° × 0.05°	daily P
ECMWF	1981–2022	Daily	0.125° × 0.125°	daily P, T, number of rain days (NRD), zonal (U) and meridional (V) winds

component analysis (PCA) Eigenvectors for temporal feature analysis. Rotated EOF (varimax rotation) was applied to improve physical interpretability. Furthermore, EOF was applied to analyze the orthogonality dynamics of the climate system in the following way:

$$Z(x,y,t) = \sum_{k-1}^{n} [PCA(t) * EOF(x,y)] \quad (1)$$

Z(x,y,t) is the original time series as a function of time (t) and space (x, y). *EOF (x, y)* shows the spatial structures (x, y) of the major factors that can account for the temporal variations of Z. *PCA(t)* is the principal component that tells you how the amplitude of each EOF varies with time. Therefore, to agree with such a type of dynamics, rotating coordinate systems is a fundamental requirement, and we applied it in this paper by rotating the old set of variables (x) to a new set of variables (z) using a projection matrix (e):

$$Z_{i=PCA} = e_{i,j} * x_{j=EOF} \quad (2)$$

The EOF analysis asks that the projection coefficients be determined in such a way that:

(1) z_1 explains the maximum possible amount of the variance of the x;
(2) z_2 explains the maximum possible amount of the remaining variance of the x;
(3) so forth for the remaining z. With these requirements, it can be shown mathematically that the projection coefficient functions (e_{ij}) are the eigenvectors of the covariance matrix of x. The fraction of the total variance explained by a particular eigenvector is equal to the ratio of that eigenvalue to the sum of all eigenvalues. It is essential to determine the redundancy of the data using the FFT model.

3. Results

3.1. Evaluation of dataset performance against in-situ data using error metrics and ANN prediction accuracy

3.1.1. Reanalysis and satellite data performance test

Evaluation of climate datasets against in-situ observations (Table 2) indicates that the ECMWF reanalysis provides the most reliable representation of P and T across the EABHA. It consistently outperforms other datasets, exhibiting the lowest errors, minimal bias, and the strongest correlation with observations. RCM shows moderate skill, with relatively low bias but higher uncertainty and weaker correlations. CHIRPS reasonably captures P magnitude but with notable uncertainty, while CRU shows consistently poor performance and the weakest agreement across all metrics.

Overall, ECMWF emerges as the most reliable dataset for characterizing climate variability in the EABHA. The ANN framework further demonstrates strong predictive capability in reproducing observed variability, confirming its robustness and suitability for altitude-dependent climate analysis and future projections. These findings are consistent with studies showing that ECMWF has reduced systematic biases and favorable RMSE compared to earlier reanalysis, and performs well for hydrological applications (Tarek et al., 2020; Hersbach et al., 2024). CHIRPS overestimated P (BR = 1.12) and showed the largest RMSE (4.02 mm), despite a moderate correlation of 0.67, which aligns with observations that CHIRPS may perform reasonably for P magnitude but exhibits regional biases (Adane et al., 2025; Chakri et al., 2025). The RCM dataset had near-zero bias (BR = 1.01) but higher RMSE (1.84 mm) and moderate correlation (0.68), reflecting the trade-offs in RCM outputs: improved spatial resolution but remaining absolute errors (EGUsphere, 2024). CRU overestimated P substantially (BR = 1.21), had moderate MRE (0.14 mm), high RMSE (2.18 mm), and the lowest correlation (0.56), indicating poor performance in capturing P patterns, consistent with studies highlighting limitations of gridded observational datasets over complex terrain (EGUsphere, 2024). Overall, ECMWF provided the most reliable representation of P, exhibiting minimal bias, low error, and strong temporal agreement with observations for both P and T, consistent with standard evaluation approaches in recent literature (Tarek et al., 2020; Hersbach et al., 2024).

3.1.2. ANN Model Prediction Performance Test

The ECMWF dataset was identified as the most reliable source for representing climate variability in the EABHA and was therefore used as input for the ANNM. This integration further confirmed its suitability for altitude-dependent climate analysis and future projections. Overall, the ANNM trained with these high-quality inputs demonstrated strong predictive capability, successfully reproducing observed variability in P, T, and NRD. Statistically significant correlations, low bias, and robust error metrics highlight the model's reliability, although a slight overestimation of temperature was noted.

The ANNM exhibited consistent and stable learning behavior across training, validation, and testing phases. The mean squared error (MSE) declined steadily over 30 epochs, reaching an optimal validation performance of approximately 1.30×10^{-8} at epoch 30. The close convergence of all learning curves indicates that the model effectively captured the underlying nonlinear relationships in the dataset without evidence of overfitting or divergence. This performance aligns with previous studies demonstrating the efficiency of ANN-based approaches in modeling complex hydroclimatic systems when trained with sufficient and well-structured datasets (Kisi and Parmar, 2016).

Error distribution analysis showed that residuals were highly concentrated around zero, particularly for the training dataset ($\approx 10^{-7}$), indicating excellent overall fit. However, a small number of validation and test samples exhibited larger deviations (10^{-4}–10^{-3}), suggesting reduced accuracy under rare or extreme conditions. This limitation is consistent with earlier findings that neural networks tend to underperform for extreme climatic events due to data scarcity and regression-to-the-mean behavior (Yaseen et al., 2015; Shamshirband et al., 2020).

Model evaluation based on scatter plots (Fig. 2) and statistical metrics (Table 3) shows a strong overall agreement between observed and simulated values. P and NRD frequency are well captured by the ANNM, indicating high predictive skill and stable performance. In contrast, T shows relatively lower accuracy, with a tendency toward higher uncertainty and slight positive bias. Overall, the model performs more reliably for P-related variables than for T, with noticeable variations

Table 2
The annual spatial mean value of precipitation (MP) in mm d^{-1} and temperature (MT) in K yr^{-1} and different error metrics include bias ratio (BR), model data per ground-based measurements, mean relative error (MRE), root mean square error (RMSE), and correlation coefficient (R) between model and ground-based measurements of precipitation and temperature data. Bracket P and T represent error metrics of precipitation in mm d^{-1} and temperature in K yr^{-1}, respectively.

Dataset: in-site: MP (0.88) &MT (291)	MP (mm d^{-1})	BR (P)	MRE (P)	RMSE (P)	R (P)	MT (K yr^{-1})	BR (T)	MRE (T)	RMSE (T)	R (T)
ECMWF	**0.89**	**0.95**	**−0.02**	**0.82**	**0.79**	**290**	**0.84**	**0.91**	**0.62**	**0.72**
CHIRPS	0.93	1.12	0.16	4.02	*0.67*	—	—	—	—	—
RCM	*0.90*	*1.01*	*0.11*	*1.84*	*0.68*	*295*	*1.06*	*0.98*	*2.65*	*0.68*
CRU	0.91	1.21	0.14	2.18	0.56	298	*0.99*	1.04	2.87	0.57

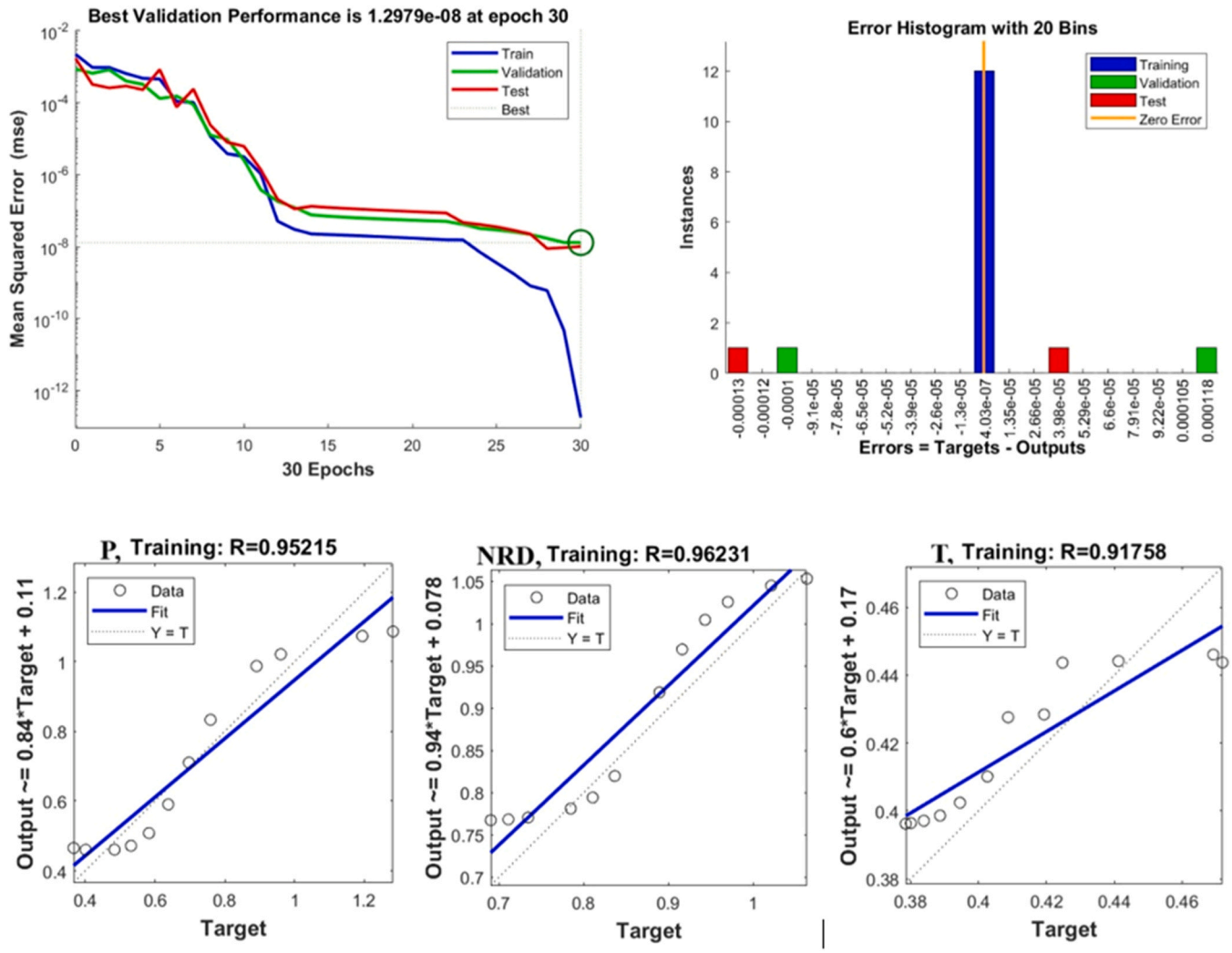


Fig. 2. Performance of the ANNM for the prediction of P, NRD, and T in the top left panel is epoch, top right panel is histogram, and bottom left, middle, and right panels are P, NRD, and T correlation coefficient, respectively.

Table 3
The accuracy of the ANNM predictions was evaluated using several error metrics: BR, MRE (mm d^{-1}), RMSE (mm d^{-1}), R, and P-value for P, T, and NRD (days yr^{-1}). These were compared between observed and predicted values at mean elevations below and above 2800 m a.s.l.

Parameter type	BR	MRE	RMSE	R	P-value
P	0.99	-0.09	0.61	0.95	0.01
NRD	1.06	0.04	0.73	0.96	0.01
T	1.11	0.12	2.82	0.92	0.02

across elevation zones above and below 2800 m a.s.l.

Regression analysis further revealed slopes slightly below unity (e.g., 0.84 for training and 0.87 for combined data), indicating a systematic tendency to underestimate high values and overestimate low values. Such bias has also been reported in previous ANN-based climate and hydrological studies. Overall, incorporating additional performance metrics such as RMSE, MAE, and Nash–Sutcliffe Efficiency (NSE) is recommended for more robust evaluation (Moriasi et al., 2007). Despite these minor limitations, the ANN model demonstrates strong capability in representing complex climate–terrain relationships, consistent with recent advances in AI-based climate modeling (Li et al., 2024).

3.2. Seasonal spatial variability of climate parameters in EABHA

During winter (DJF), the historical climatology (1981–2022; Fig. 3a–b) exhibits a well-organized large-scale circulation dominated by winter monsoon dynamics. The wind field shows a coherent and stable structure, with zonal wind (U) ranging from −5 to 1 m s^{-1} (mean: −2.1 m s^{-1}) and meridional wind (V) varying from −5 to 4 m s^{-1} (mean: −1.5 m s^{-1}), indicating dominant east–west and north–south flow components. These patterns collectively reflect a prevailing northwesterly to northeasterly circulation, consistent with the advection of cold and dry continental air masses from higher latitudes. P is generally minimal across the domain, with near-zero values (~0.5 mm d^{-1}) mainly confined to the southeastern sector, while most regions remain dry. This dryness is sustained by moisture-limited continental airflow influenced by hemispheric land–ocean asymmetry, where the majority of oceanic moisture sources lie in the Southern Hemisphere (Megbar et al., 2019). T ranges from 280 to 300 K yr^{-1}, with a mean of 295 K yr^{-1}, exhibiting a clear meridional gradient characterized by colder northern and relatively warmer southern conditions. Overall, the DJF circulation reflects a stable winter monsoon regime with strong continental influence, organized wind fields, and limited synoptic variability.

Future projections (2023–2050; Fig. 3b) indicate a marked weakening of winter monsoon dynamics. Both U and V components decrease in magnitude, reflecting a reduction in large-scale pressure gradients

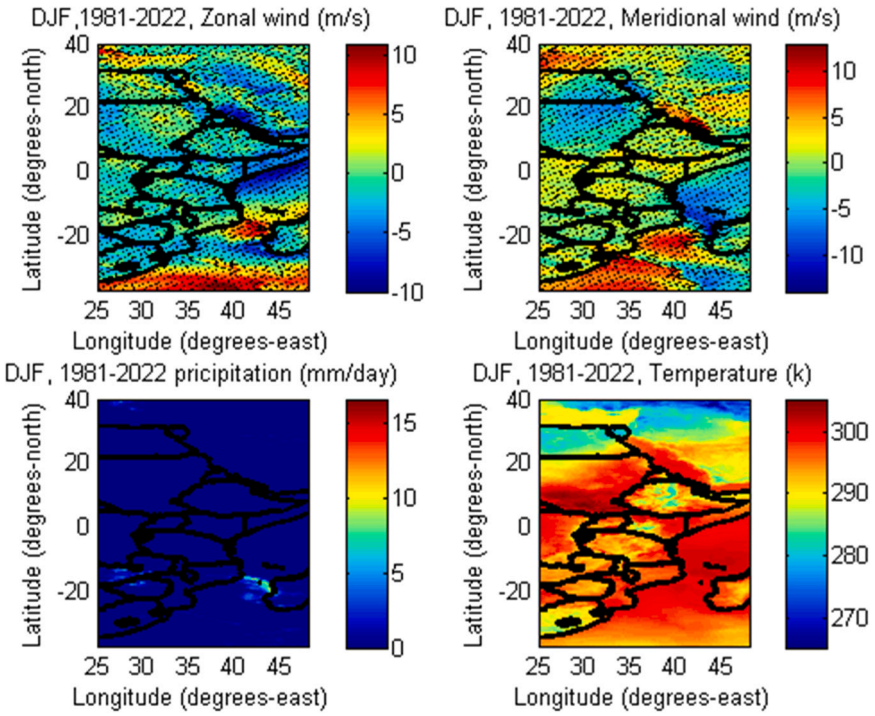


Fig. 3a. During December, January, and February (DJF) (winter), spatial variability of U, left top panel, V, right top panel; P, left bottom panel, and T, right bottom panel. The black arrow indicates the wind direction with the historical data from 1981 to 2022.

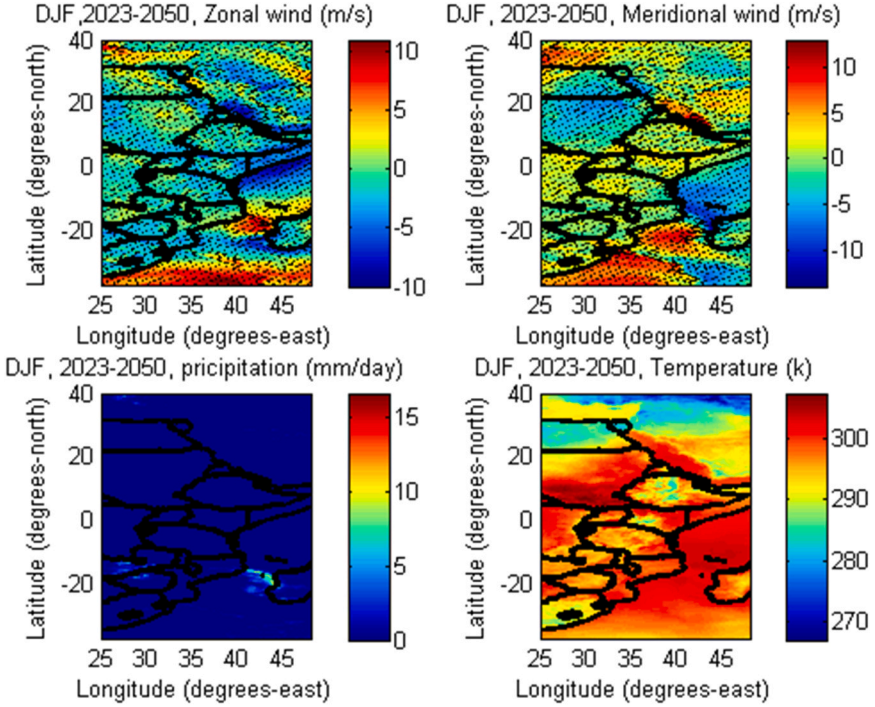


Fig. 3b. During DJF, spatial variability prediction of U, V, P, and T from 2023 to 2050.

and weakened cold-air advection into the region. P and pressure gradients become increasingly homogenized, suggesting a more thermodynamically uniform atmospheric state. T shows a consistent and widespread increase, accompanied by a weakened north–south thermal gradient due to stronger warming in northern regions. Collectively, these changes indicate a transition toward a warmer and dynamically weaker winter regime, characterized by reduced circulation intensity and a less distinct monsoonal structure under future climate conditions.

During spring (MAM), the historical atmospheric state (1981–2022; Fig. 4a) represents a transitional phase between winter and summer monsoon systems. U ranges from −5 to −1.5 m s^{-1} (mean: −2.2 m s^{-1}), while V varies from −5 to 2 m s^{-1} (mean: 0.5 m s^{-1}), indicating a gradual shift toward southerly flow and weakening winter monsoon influence. This transition enhances moisture transport from the Indian Ocean

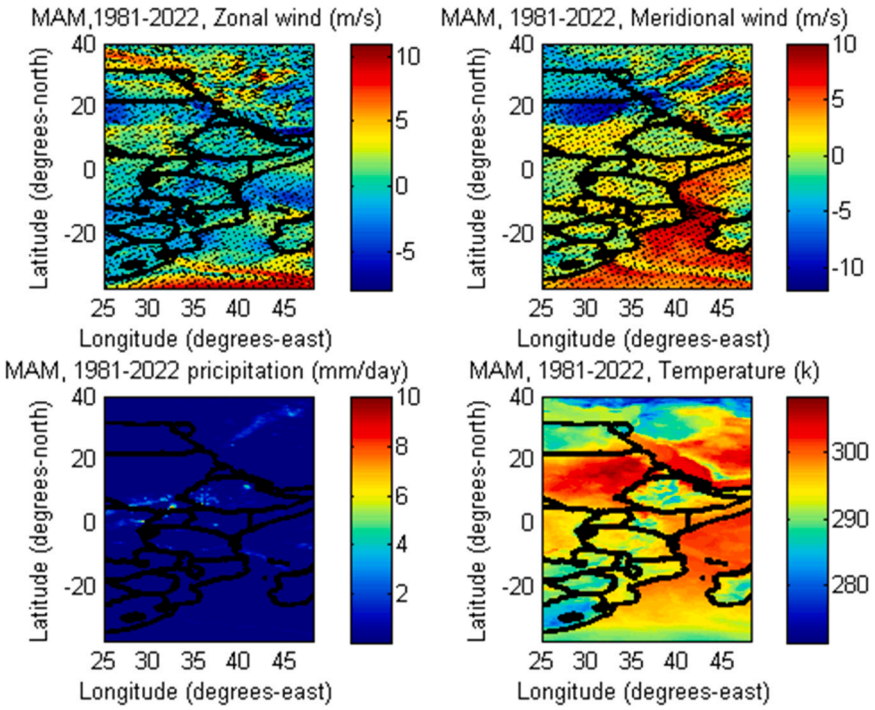


Fig. 4a. During March, April, and May (MAM) (spring), historical data spatial variability of U left top panel, V right top panel, P left bottom panel, and T right bottom panel from 1981 to 2022.

toward eastern and central parts of the EABHA, contributing to increasing P. T ranges from 280 to 298 K yr^{-1} (mean: 292 K yr^{-1}) and shows a consistent warming trend driven by increasing solar insolation and land-surface heating. The evolving pressure gradient associated with seasonal heating reinforces zonal flow and promotes atmospheric instability, leading to intensified convection and gradually increasing P. Consequently, the region transitions into the pre-monsoon phase, characterized by enhanced meridional exchange and increasing atmospheric variability (Megbar et al., 2019).

Projected conditions (2023–2050; Fig. 4b) indicate an accelerated and more pronounced transition toward southwesterly circulation, suggesting an earlier onset of monsoon-like dynamics. Strengthening of both U and V components reflects intensified low-level convergence and enhanced dynamical forcing. P is projected to increase in both magnitude and spatial concentration, particularly over central and southern regions, indicating a heterogeneous amplification of pre-monsoonal

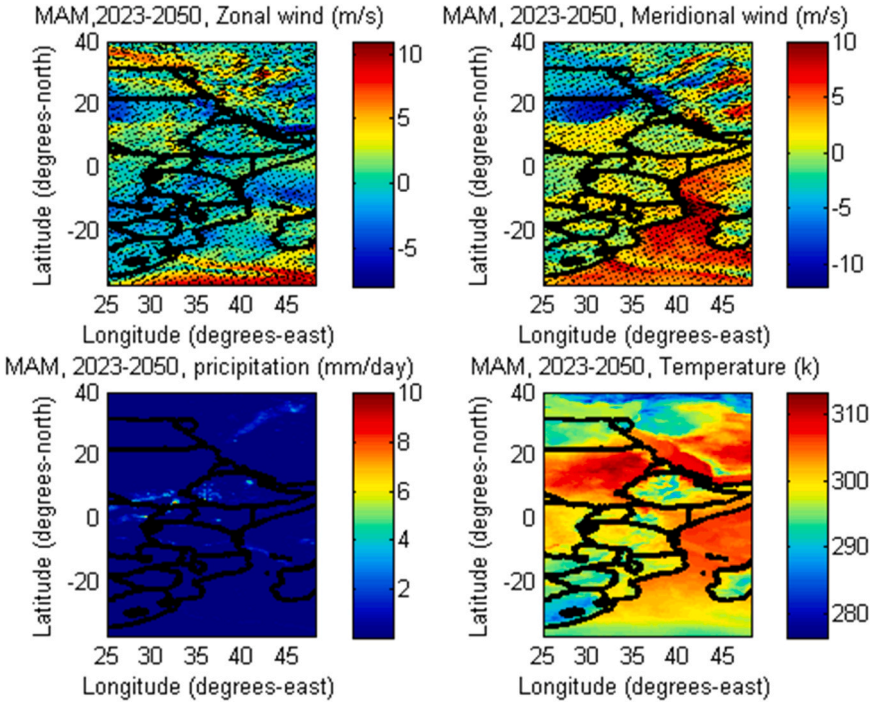


Fig. 4b. During MAM, spatial variability prediction of U, V, P, and T in the EABHA from 2023 to 2050.

rainfall. T continues to rise strongly, with greater warming over land areas driven by land–atmosphere feedback mechanisms. Overall, MAM evolves toward a warmer, wetter, and dynamically more active season, characterized by an earlier monsoon transition and intensified convective processes.

During summer (JJA), the historical circulation (1981–2022; Fig. 5a) is dominated by a strong monsoonal flow that efficiently transports moisture into the continental interior. U ranges from −5 to 4 m s^{-1} (mean: 2.1 m s^{-1}), while V varies from 0 to 3 m s^{-1} (mean: 1.9 m s^{-1}), reflecting a dominant southwesterly flow regime. P ranges from 0 to 2 mm d^{-1} (mean: 1.8 mm d^{-1}), while T varies between 275 and 290 K yr^{-1} (mean: 287 K yr^{-1}). Moisture transport from oceanic source regions supports widespread rainfall across the domain, with spatial variability primarily controlled by topography and convective processes (Finch et al., 2011). Despite high seasonal T, persistent cloud cover, and moderate diurnal T extremes, rainfall reduces thermal variability across the region.

Future projections (2023–2050; Fig. 5b) suggest a slight weakening and spatial redistribution of monsoon circulation. Reduced wind intensity indicates weakened pressure gradients and less coherent large-scale forcing in certain regions. P becomes increasingly heterogeneous, with localized increases and decreases, indicating a shift from spatially uniform monsoonal rainfall toward a more variable precipitation regime. T shows a persistent and robust increase across the domain, including during peak monsoon months, indicating dominance of background warming over seasonal variability. Overall, JJA is characterized by sustained warming, slightly weakened monsoon circulation, and increased precipitation variability, potentially enhancing localized extreme rainfall events.

During autumn (SON), the historical circulation (1981–2022; Fig. 6a) reflects the monsoon withdrawal phase and transition toward dry-season conditions. U ranges from −5 to 0 m s^{-1} (mean: −1.7 m s^{-1}), while V varies from −5 to 2 m s^{-1} (mean: −2.3 m s^{-1}), indicating a gradual shift toward northeasterly flow. P is very low (0–2 mm d^{-1}; mean: 0.1 mm d^{-1}), and T averages 284 K yr^{-1}. The seasonal transition is driven by the southward migration of solar heating and the establishment of continental high-pressure systems, leading to reduced moisture availability and declining rainfall. Residual warmth persists, particularly in southern regions, due to delayed radiative cooling, even as overall T declines after the summer peak (Finch et al., 2011; Megbar et al., 2019).

Future projections (2023–2050; Fig. 6b) indicate a delayed and weakened monsoon withdrawal. Northeasterly circulation develops later and more gradually, extending the transition between wet and dry seasons. P persists further into SON, suggesting an extension of the wet season and a delayed onset of dry conditions. T remains anomalously high, particularly in late SON, slowing seasonal cooling and prolonging warm conditions. Overall, SON is projected to experience delayed monsoon retreat, extended P, and sustained warming, indicating a lengthened and thermally intensified transitional season under future climate change.

3.3. *Temporal variability and prediction of P, T, and NRD under altitude variation*

The results (Table 4) indicate a consistent warming signal across the EABHA for both elevation zones, with stronger and more pronounced T increases at higher elevations (≥2800 m a.s.l.). During the historical period (1981–2022), T trends were positive in both zones, but the rate was notably higher in the highlands (0.030 K yr^{-1}) compared to the lowlands (0.002 K yr^{-1}). This elevational contrast intensifies in the projection period (2023–2050), where warming accelerates further, particularly in the highlands (0.040 K yr^{-1}). P trends show a general decline during the historical period in both zones, with a sharper decrease in the lowlands. However, future projections suggest a divergence: P continues to decrease in the highlands (−0.030 mm d^{-1}), while a slight recovery or increase is projected in the lowlands (0.013 mm d^{-1}), indicating increasing spatial variability in rainfall patterns.

Similarly, the NRD exhibits a consistent declining trend across both elevation ranges and periods (Table 5). Historically, the reduction in NRD is more pronounced in the highlands (−0.027 days yr^{-1}) than in the lowlands (−0.005 days yr^{-1}), suggesting a shift toward fewer rainy days

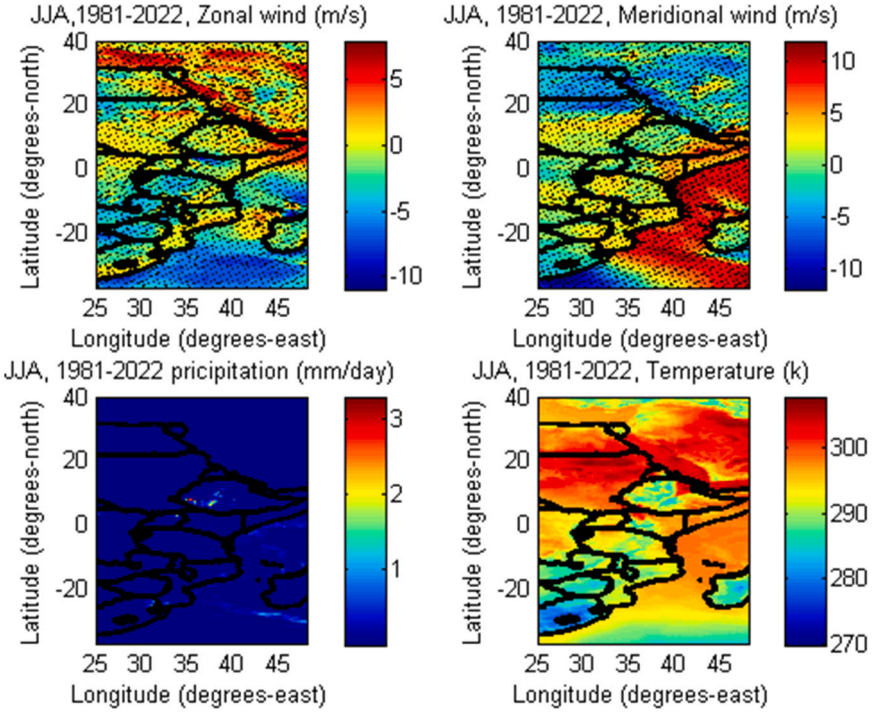


Fig. 5a. June, July, and August (JJA) (summer) historical data in EABHA spatial variability of U left top panel, V right top panel, P left bottom panel, and T right bottom panel from 1981 to 2022.

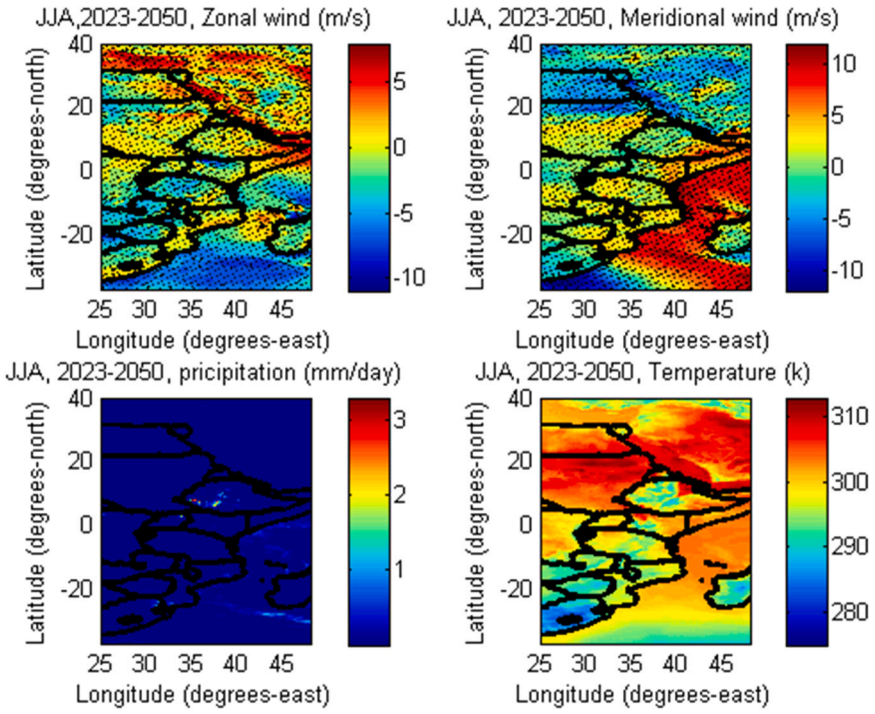


Fig. 5b. JJA the spatial variability prediction of U, V, P, and T from 2023 to 2050.

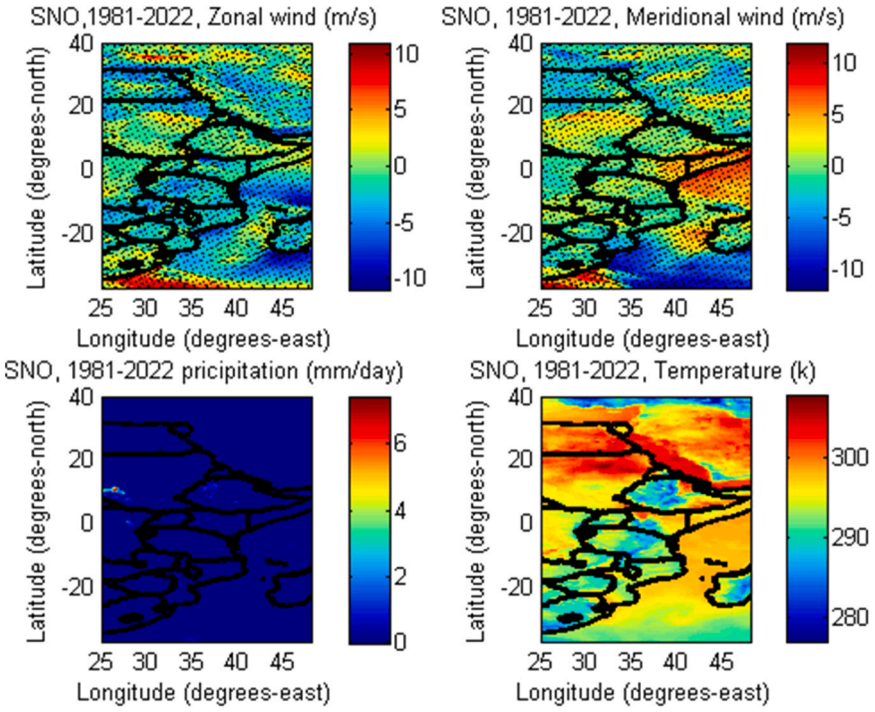


Fig. 6a. September, October, and November (SON) (autumn) historical data spatial variability in EABHA of U left top panel, V right top panel, P left bottom panel, and T right bottom panel from 1981 to 2022.

at higher elevations. This trend intensifies in the future (2023–2050), with both zones experiencing stronger declines, particularly in the lowlands (−0.047 days yr^{-1}). Overall, the combined patterns of increasing T, decreasing or variable P, and declining NRD suggest a transition toward warmer and potentially drier conditions, with heightened climate variability and elevation-dependent impacts across the region.

The temporal variability of P and T in the study area is presented in Fig. 7. Analysis shows that annual P has remained relatively stable over the past 40 years and is predicted to continue this pattern over the next three decades. In contrast, the NRD has steadily declined over the same period and is expected to continue decreasing. Annual mean T above 2800 m a.s.l. areas, exhibits a clear upward trend, with the most pronounced increases ranging from −0.310–1.940 K yr^{-1} with an average value of 0.040 K yr^{-1} projected for 2050 years, while P trends decrease slowly, appearing nearly constant over time (top panel, Fig. 7). Between

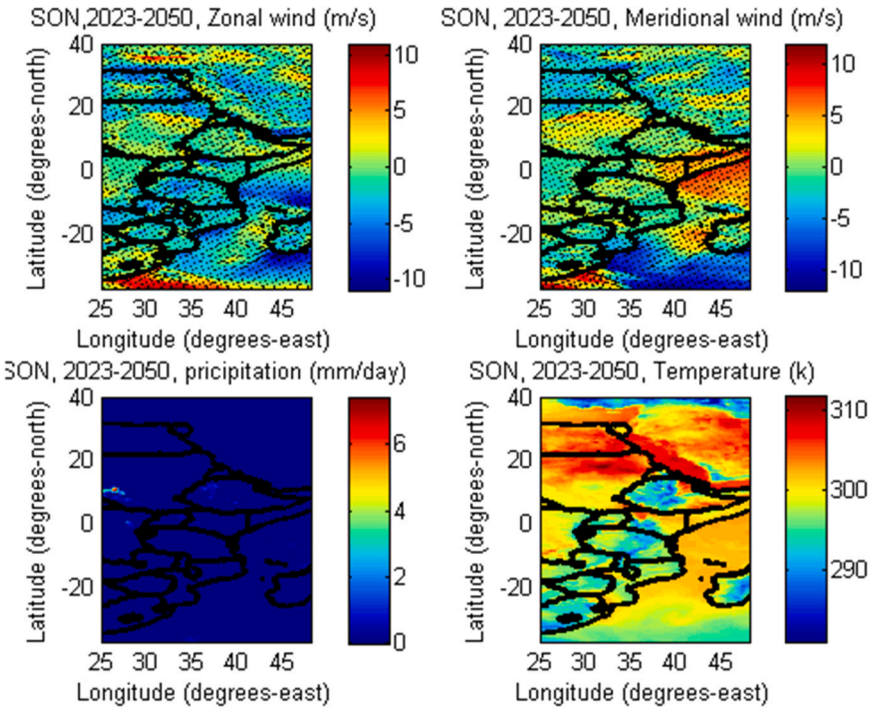


Fig. 6b. SON spatial variability prediction data of U, V, P, and T from 2023 to 2050.

Table 4
Minimum, maximum, and trend rate of P (mm d^{-1}) and T (K yr^{-1}) above and below 2800 m a.s.l. in EABHA (95% CI, ANNM-based estimates).

Parameter	Period	Min (≥2800 m a.s.l.)	Max (≥2800 m a.s.l.)	Rate (≥2800 m a.s.l.)	Decision support	Min (<2800 m a.s.l.)	Max (<2800 m a.s.l.)	Rate (<2800 m a.s.l.)	Decision support
T	1981–2022	-1.900	1.000	0.030	Increase	-2.080	1.230	0.002	Increase
T	2023–2050	-0.310	1.940	0.040	Increase	-2.300	2.450	0.005	Increase
P	1981–2022	-2.060	2.600	-0.010	Decrease	-1.900	1.609	-0.073	Decrease
P	2023–2050	-1.560	0.710	-0.030	**Decrease**	-1.613	2.012	0.013	**Increase**

Table 5
Minimum, maximum, and trend rate of NRD (days yr^{-1}) above and below 2800 m a.s.l. in the eastern Afromontane hotspot area (95% CI, ANNM-based estimates).

Parameter	Period	Min (≥2800 m a.s.l.)	Max (≥2800 m a.s.l.)	Rate (≥2800 m a.s.l.)	Decision support	Min (<2800 m a.s.l.)	Max (<2800 m a.s.l.)	Rate (<2800 m a.s.l.)	Decision support
NRD	1981–2022	-1.607	2.706	-0.027	Decrease	-1.239	1.451	-0.005	Decrease
NRD	2023–2050	-1.498	0.515	-0.033	Decrease	-1.987	0.672	-0.047	Decrease

1981 and 2022, T at high elevations remained relatively stable or slightly negative; however, after 2010, a sharp increase is observed. In lower-altitude regions (<2800 m a.s.l.), T showed modest positive trends from 1981 to 2022, followed by stronger fluctuations between 2023 and 2050. The daily mean of annual P trends fluctuated between −1.60 and 2.12 mm d^{-1} , with minimal change below 2800 m a.s.l. and a slight decline above 2800 m a.s.l. (Fig. 7, Tables 4 and 5).

The NRDs declined faster at higher elevations than at lower altitudes. Above 2800 m a.s.l., from 1981 to 2022, P decreased by −0.010 mm d^{-1} , whereas NRDs declined more rapidly at −0.027 days yr^{-1}. Predictions for 2023–2050 suggest further reductions, with P decreasing by −0.030 mm d^{-1} and NRD by −0.033 days yr^{-1}. Below 2800 m a.s.l., P slightly decreased (-0.073 mm d^{-1} and NRD remained nearly constant (-0.005 days yr^{-1}) from 1981 to 2022. Between 2023–2050, P is predicted to increase slightly (0.013 mm d^{-1}), while NRDs continue to decline (-0.047 days yr^{-1}) ($p < 0.05$). Overall, P remains relatively stable; however, warming T is accompanied by a consistent reduction in NRDs, particularly at higher elevations, which corroborates earlier findings (IPCC, 2007).

Figs. 7 and 8, together with Tables 4 and 5, illustrate the altitude-dependent temporal trends of T, P, and the NRD across the EABHA from 1981 to 2022, and projected from 2023 to 2050. At elevations above 2800 m a.s.l., T has shown a marked increase over the past four decades, with trends rising from −1.90–1.00 K yr^{-1} (1981–2022) at a rate of 0.03 K yr^{-1} , and predictions for 2023–2050 indicate further acceleration to 1.94 K yr^{-1} (Table 3; Figs. 7–8). In contrast, P remains relatively stable but exhibits a slight declining trend (-0.010 mm d^{-1} historically, −0.030 mm d^{-1} predicted), while the frequency of NRD has decreased more substantially (-0.027 days yr^{-1} historically, −0.033 days yr^{-1} predicted), suggesting fewer but moderately consistent rainfall events at high altitudes.

At lower elevations (<2800 m a.s.l.), T trends are more moderate, ranging from −2.08–1.23 K yr^{-1} historically and −2.30–2.45 K yr^{-1} predicted, with rates of 0.002 K yr^{-1} (1981–2022) to 0.005 K yr^{-1} (2023–2050) (see Table; Fig. 7 & 8). P has shown minor historical decreases (-0.073 mm d^{-1}) but is predicted to slightly increase (0.013 mm d^{-1}) in the future, while the NRD decline gradually (-0.005 days yr^{-1} historically, −0.047 days yr^{-1} predicted). These patterns

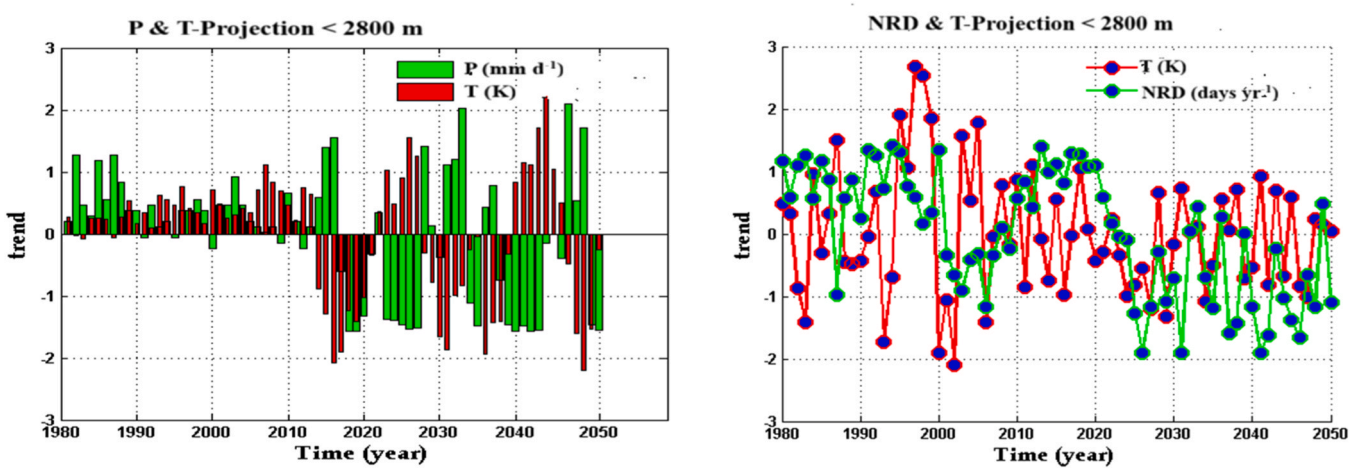


Fig. 7. The daily mean of the annual trend of P (mm d^{-1}) with respect to annual T (K yr^{-1}) (left panel), and NRD (days yr^{-1}) with respect to T (K yr^{-1}) (right panel), for areas below 2800 m a.s.l. The trends are calculated relative to a reference year of 1980 and a baseline period of 1981–2022 for the period 2023–2050 in EABHA.

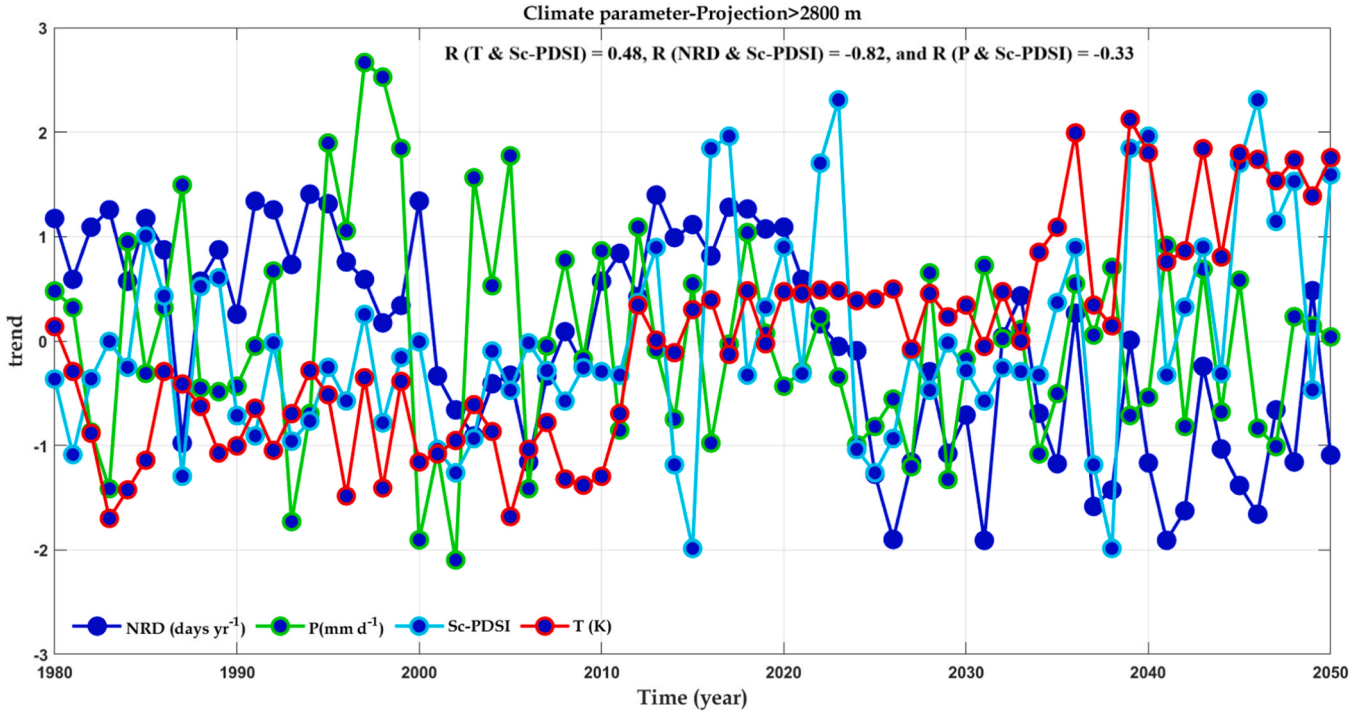


Fig. 8. Comparisons of the P, T, and NRD with respect to Sc-PDSI in EABHA at an altitude above 2800 m a.s.l.

indicate that lower-altitude regions experience slower warming, more stable rainfall totals, but a gradual reduction in rainy days, consistent with broader regional climate projections. Overall, the combined analysis highlights three key points:

1. Accelerated warming at high elevations, with the most significant temperature increases above 2800 m a.s.l.
2. Relatively stable precipitation totals, yet a declining number of rain days, particularly at higher altitudes, which may reduce effective moisture availability despite consistent rainfall amounts.
3. Elevation-dependent climate vulnerability, suggesting that alpine ecosystems are more susceptible to the combined impacts of warming and altered rainfall patterns, with implications for biodiversity, hydrology, and livelihoods.

These trends underscore the importance of altitude-specific climate adaptation strategies for sustainable ecosystem and resource management in the EABHA and are in agreement with previous studies related to drought impacts on tropical highland climate change (IPCC, 2007; Megbar et al., 2019).

Analysis of the 1981–2022 historical and projected from 2022 to 2050 hydroclimatic records indicates a clear elevation-dependent response of hydroclimate and drought in the EABHA. Below 2800 m a.s.l., warming is associated with declining daily mean P and a reduction in the NRD (Fig. 7), producing longer dry spells and reduced effective recharge for soils and streams. Above 2800 m a.s.l., correlation analysis with the Sc-PDSI (Fig. 8) reveals contrasting controls: T correlates positively with Sc-PDSI (R = +0.48), while the NRDs and total P show strong negative correlations with Sc-PDSI (R = –0.82 and R = –0.33, respectively). Together, these coefficients indicate that (1) declines in NRD exert the strongest influence on worsening Sc-PDSI (more negative values), (2) decreases in total P also contribute to drought but with a

weaker effect than NRD, and (3) the positive T–Sc-PDSI association at high elevations suggests that T variations are co-varying with processes that can locally increase water availability (Fig. 8). The combined trends imply intensified drought risk across elevations, driven mainly by reductions in rainfall frequency and, secondarily, by reduced rainfall totals and enhanced evaporative demand.

4. Discussion

ECMWF P and T data showed the strongest agreement with in-situ measurements, characterized by the lowest MRE, minimal RMSE, and the highest R. These findings align with recent studies showing that ERA5 and other ECMWF products outperform CHIRPS, CRU, and regional climate models in complex topographies such as the Ethiopian highlands (Hersbach et al., 2020; Ojara et al., 2023). CHIRPS displayed relatively higher MRE for P, consistent with multiple assessments across East Africa where CHIRPS tends to underestimate convective rainfall and overestimate light rains (Gebrechorkos et al., 2019; Dinku, 2020). The performance ranking of climate datasets in this study is consistent with previous evaluations over the Upper Blue Nile Basin (UBNB). The superior performance of the ECMWF reanalysis agrees with Megbar et al. (2019) and Tarek et al. (2020), who reported that reanalysis products generally outperform satellite-based and gridded observational datasets in reproducing P and T variability across the basin, particularly when validated against sparse in-situ observations. This improved performance is attributed to advanced data assimilation frameworks and bias correction schemes that better capture both spatial and temporal climate variability in complex highland terrain.

Similarly, the strong agreement found for ECMWF is consistent with the latest generation reanalysis described by Hersbach et al. (2024), which demonstrates improved representation of atmospheric processes, enhanced spatial resolution, and reduced systematic bias compared to earlier reanalysis products. However, despite these improvements, previous studies also note persistent uncertainties in P estimation over complex topography, which is consistent with the moderate performance of RCM and CHIRPS in the present study. In line with findings from regional assessments, CHIRPS generally performs better for rainfall magnitude but remains less accurate in capturing variability at higher elevations, while CRU typically shows lower skill due to interpolation-based limitations. Overall, the agreement between the present results and previous studies confirms that reanalysis products, particularly ECMWF, provide the most robust baseline for climate characterization in the UBNB, while satellite- and observation-based products introduce varying degrees of uncertainty depending on topography and variable type.

The spatial analysis of wind circulation reveals pronounced seasonal shifts in the U and V components, driven by hemispheric pressure gradients. In winter, negative U and V indicate dominant easterly and northerly flows linked to the Arabian high-pressure system and limited northern moisture input (Nicholson, 2017; Megbar et al., 2019), resulting in very low P, especially in the southeast. In spring, negative U and positive V reflect stronger easterly and southerly moisture transport from the Indian Ocean, consistent with the influence of the Indian Ocean Walker circulation (Saji and Yamagata, 2023). Summer brings strong positive U and V, marking the onset of the West African monsoon and Congo Air Boundary, which enhances rainfall, aligning with previous observations of monsoon strengthening (Finch et al., 2011; Nicholson, 2017; Diem et al., 2022). By autumn, weakening transport and negative U and V correspond to monsoon retreat and renewed subtropical highs, producing dry conditions (Megbar et al., 2019). Long-term projections show clear warming across the eastern Afromontane region, with temperatures rising about 0.040 K yr^{-1} , roughly 1.20 K per three decades, especially above 2800 m a.s.l., consistent with elevation-dependent warming (IPCC, 2021; Pepin et al., 2022), which heightens vulnerability in montane ecosystems.

The ANNM demonstrated strong capability in predicting P and NRD, with BR near unity and high correlations (R = 0.95–0.96), indicating that the model effectively captures both rainfall magnitude and occurrence across elevation classes. This finding is consistent with recent advances in machine-learning modeling for monthly P prediction (El Hafyani et al., 2024) and with the use of ANN- and neuro–fuzzy–based rainfall models in Ethiopia, which have produced similarly robust skills when appropriately trained with geographic inputs (Abebe and Endalie, 2023). A broader review by Oliveira et al. (2023) also emphasizes that machine-learning models outperform conventional statistical approaches in hydro-climatic prediction when nonlinear controls dominate, supporting the strong ANN behaviour observed here. In contrast, the T in predictions show larger uncertainty, with a warm bias (BR ≈ 1.11; MRE ≈ 0.12 K yr^{-1}) and higher RMSE (∼ 2.82 K yr^{-1}), despite maintaining a statistically significant correlation (R = 0.92). This pattern agrees with results from mountainous regions, where T is more sensitive to elevation-dependent processes. A recent intercomparison of ANNM, Random Forest, and CNN downscaling of near-surface T over the Alps demonstrated (Mulualem et al., 2024). Similarly, Mulualem et al. (2024) reported that climate forecasting models over the Upper Blue Nile Basin experience increased T errors at high altitude due to steep lapse-rate variability, local microclimates, and sparse station density. Comparable findings from Himalayan altitudinal gradient modelling also show that T and P prediction errors increase where terrain complexity is high and data coverage is uneven (Ntirugulirwa et al., 2023).

Given these challenges, recent literature recommends several pathways for improvement, including the integration of topographic predictors (Mulualem et al., 2024), the use of hybrid or deep sequence architectures such as CNN-LSTM (Oliveira et al., 2023), or the application of bias-correction procedures to address systematic warm biases (Ntirugulirwa et al., 2023). Overall, the ANNM results reported here are well aligned with contemporary machine-learning findings: P variables are predicted with high fidelity, while T prediction remains more challenging in mountainous environments due to unresolved fine-scale topographic effects.

P trends display strong year-to-year variability but remain broadly stable over the long term, consistent with evidence that East African rainfall is unlikely to decline sharply even as its spatial patterns shift (Wainwright et al., 2021; Lyon et al., 2023). However, the marked reduction in NRD frequency, despite steady annual rainfall, indicates a shift toward fewer but more intense events, a trend increasingly noted in regional and global assessments (IPCC, 2021; Otto et al., 2023), with implications for flooding, groundwater recharge, and soil moisture deficits. P declines more rapidly above 2800 m a.s.l., and NRD reductions occur across all elevations, raising concerns that montane ecosystems may face heightened drought risk despite stable mean rainfall, as earlier work also suggested (Marchant and Hooghiemstra, 2004; Burgess et al., 2007). These shifts threaten the eastern Afromontane hotspot's climate-dependent biodiversity (Burgess et al., 2007), as accelerated warming and fewer rainy days reduce soil moisture, increase evapotranspiration, disrupt streamflow, and intensify risks for species with limited elevational ranges (Fig. 8) (Carlson et al., 2022; Gizaw et al., 2023) as well as for millions relying on rain-fed agriculture (World Bank, 2020). Although current climate stability aligns with past reconstructions (Marchant and Hooghiemstra, 2004), projected warming and altered rainfall frequency support warnings of growing climate extremes in East African highlands (IPCC, 2021), and the divergence between stable annual totals and declining NRD frequency echoes findings of Wilfred (2010) and recent modelling studies (Lyon et al., 2023), suggesting that future ecological impacts will be driven more by changes in rainfall timing and intensity than by total seasonal rainfall.

The observed declines in P and NRD frequency below 2800 m a.s.l signal increasing hydroclimatic stress for rain-fed agriculture and ecosystems in the EABHA, consistent with worsening drought trends across Ethiopian basins (Wubneh et al., 2023; Mulualem et al., 2024). Reduced

NRD strongly amplifies drought severity by lowering soil moisture recharge and extending dry spells, as reflected in the very strong negative correlation with Sc-PDSI (R = –0.82) (Fig. 8). P shows a moderate negative correlation with Sc-PDSI (R = –0.33), indicating that rainfall distribution and timing often outweigh amounts in determining drought impacts in complex terrain. At elevations > 2800 m a.s.l., the moderate positive correlation between T and Sc-PDSI (R = +0.48) likely reflects orographic effects and interactions between warming and moisture-bearing circulations, where higher T can increase atmospheric moisture, enhance infiltration, or reduce snow/soil freeze (Fig. 8). However, increased potential evapotranspiration (PET) can amplify drought if not matched by P, highlighting PET/ET sensitivity in drought indices (Yang et al., 2020). Overall, drought in the EABHA exhibits elevation-dependent pathways: low and mid-elevations are mainly constrained by declines in both rainfall and NRD frequency, while high elevations are influenced by rainfall timing and temperature–moisture interactions mediated by topography, underscoring the importance of rainfall frequency, PET formulation, and topography in regional drought risk assessments (Marra et al., 2022).

Careful selection, validation, and bias correction of climate datasets remain critical for climate research in complex mountain regions (Gebrechorkos et al., 2020; Harrigan et al., 2020). High-quality climate information supports SDG 13 (Climate Action) by improving the reliability of climate-resilience planning (IPCC, 2022). In the studied highland environment, seasonal U–V wind patterns reveal strong hemispheric pressure-gradient influences, with winter dominated by easterly and northerly flow and spring–summer transitions enabling moisture transport from the Indian Ocean and Atlantic sectors; these circulation dynamics are consistent with broader East African climatology and are central to understanding future regional climate risks (Nicholson, 2017).

Accurate P estimates also underpin SDG 2 (Zero Hunger). Although annual rainfall totals may not change substantially in some projections, the decline in NRDs frequency suggests a shift toward more intense, episodic rainfall events (Funk et al., 2015; Fowler et al., 2021). Such changes reduce soil moisture retention, limit groundwater recharge, increase surface runoff, and ultimately threaten the productivity of rain-fed agricultural systems (Thornton et al., 2014). These findings align with regional evidence showing that altered rainfall timing and intensity are becoming primary constraints for agricultural resilience (Funk et al., 2015).

Shifts in wind-driven moisture transport and rainfall seasonality further affect SDG 6 (Clean Water and Sanitation). Montane watersheds rely on steady, well-distributed rainfall to support recharge and streamflow stability; increasing rainfall intensity and decreasing event frequency weaken water security, reduce infiltration efficiency, and exacerbate hydrological variability (Conway et al., 2018). Improved understanding of these processes is therefore essential for sustainable water-resource planning under accelerating climatic change (IPCC, 2022).

The most pronounced warming occurs above roughly 2800 m a.s.l., aligning with well-established elevation-dependent warming in mountain regions (Pepin et al., 2015). Recent studies from East African ranges, including Mount Kenya, show similarly rapid temperature increases in both observational and reanalysis records (Olago and Nyumba, 2023), threatening species with restricted thermal tolerances and limited room to shift upslope. These patterns heighten extinction risks and underscore key vulnerabilities related to SDG 15 (Life on Land) (Williams et al., 2021).

Across these sectors, rising T and altered P regimes create interconnected risks. Strengthening ecosystem resilience, enhancing water harvesting and storage, and scaling climate-smart agriculture provide cross-cutting benefits that support SDG 2, SDG 6, SDG 13, and SDG 15 (IPCC, 2022; Chirwa et al., 2020). To advance SDG-aligned adaptation, three priority areas emerge: (1) expansion of in-situ climate monitoring networks in high-elevation terrain to better capture elevation-dependent warming (Pepin et al., 2015; Olago and Nyumba, 2023); (2) continued validation and bias correction of reanalysis and satellite datasets for mountain environments (Gebrechorkos et al., 2020; Harrigan et al., 2020); and (3) integrated assessments linking shifts in NRDs frequency to ecological function, hydrological behavior, and agricultural productivity (Conway et al., 2018; Fowler et al., 2021). These efforts will facilitate evidence-based adaptation decisions and strengthen the climate resilience of high-elevation socio-ecological systems (IPCC, 2022).

4.1. *Discussion under different scenarios*

The projection of P, T, and NRD in the EABHA reveals strong and elevation-dependent climate signals, with clear contrasts between areas above and below 2800 m a.s.l. Across all scenarios, a consistent warming trend emerges from 1981 to 2050, accompanied by a progressive decline in NRD, indicating a shift toward fewer but more intense rainfall events. However, the magnitude of change varies markedly depending on the interaction between climate forcing, land-use dynamics, and population pressure. Under the combined climate, land-use, and population change Scenario, hydro-climatic instability intensifies, with stronger warming at high elevations (up to 0.040 K yr^{-1}), accelerating drying above 2800 m a.s.l., and increasingly erratic rainfall at lower elevations driven by enhanced surface heating and land conversion. The severe climate change Scenario amplifies these signals further, producing even higher warming rates (potentially >0.04–0.05 K yr^{-1}), sharper P divergence between elevations, longer dry spells, and more extreme convective storms. In contrast, smoother and more moderate trajectories Scenario, where warming is steady but less disruptive, P changes remain limited, and NRD declines gradually without abrupt shifts, reflecting a more stable system when land-use and demographic pressures are held constant.

The consistently declining NRD across all scenarios signals a transition toward more concentrated rainfall extremes, increasing risks of floods, droughts, and soil erosion. When compared with standard RCP/SSP frameworks (O'Neill et al., 2016; IPCC, 2021), Scenario 3 broadly aligns with moderate pathways (e.g., SSP2–4.5), whereas Scenarios 1 and 2 reveal substantially stronger regional amplification due to land-surface and elevation feedbacks that are not fully captured in global models. This divergence highlights a key insight: while RCP/SSP scenarios emphasize emission-driven global change, the EABHA projections underscore local sensitivity to land transformation and topography, with Scenarios suggesting that real-world impacts may exceed global projections. Consequently, integrating climate dynamics with land-use and population change is essential for accurately capturing future hydro-climatic risks in complex mountainous systems.

5. Conclusions and recommendations

In the EABHA, P has remained relatively stable over the past four decades and is projected to continue, while T is above 2800 m a.s.l. shows pronounced warming consistent with elevation-dependent trends. Despite stable rainfall, the number of NRD is steadily declining, particularly at high elevations. This reduction in P frequency, combined with rising T, increases hydroclimatic stress and drought risk, with NRD decline strongly linked to drought severity, while total P plays a secondary role. Future warming and variable P patterns threaten high-elevation biodiversity, agriculture, and livelihoods, with implications for SDG 2 (Zero Hunger), SDG 6 (Clean Water), SDG 13 (Climate Action), and SDG 15 (Life on Land). Adaptation strategies should be altitude-specific in agriculture, adaptive planting, soil moisture conservation, and seasonal forecasting. Policy and planning must integrate local climate projections, strengthen institutional capacity, and support community-based adaptation. Overall, rising T and declining NRD underscore the need for proactive, elevation-sensitive climate action to safeguard ecosystems, water resources, and livelihoods in the EABHA.

CRediT authorship contribution statement

Teferi Minbiyew: Writing – original draft, Methodology, Investigation, Data curation. **Birhan Megbar Wondie:** Writing – review & editing, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

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Declaration of Competing Interest

The authors declare that there is no conflict of interest regarding the publication of this paper.

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Data availability

The data that has been used is confidential.

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